

EFFECTIVE-TOGGLE PERSISTENCE FOR ROLLING FORECASTS OF SURFACE-CODE LOGICAL ERROR RATES

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Abstract. Logical error rates accumulated over repeated error-correction rounds are bounded probabilities with a known parity geometry. A stationary per-round toggle model respects that geometry but fails on the public Google surface-code hardware record: the effective toggle inferred from measured rates changes systematically with circuit depth. This paper introduces effective-toggle persistence, a rolling forecast that inverts the most recent rate within each hardware stratum and propagates that effective toggle analytically to later rounds. The method requires no numerical optimization. The analysis proves exactness under stationary toggles, an r -Lipschitz error bound under drift, and $O(N + G)$ work with $O(G)$ storage for N records and G strata. The protocol uses 130 hardware experiments, 50,000 shots per experiment, two code distances, two logical bases, and thirteen round settings. Tensor-network decoder rates define development; belief-matching decoder rates are reserved as a distinct confirmation endpoint. Across four rolling folds, the proposed forecast reduces mean-squared error relative to the strongest of a stationary toggle fit, nearest-round rate, and ridge extrapolation by 35.8–61.2 percent on development and 13.0–58.4 percent on confirmation. It wins every fold. The measured implementation is over three orders of magnitude faster than the registered grid fit at ten strata, while the nearest-rate baseline remains faster and substantially less accurate. The result is a hardware-record forecasting statement, not a stationary noise-model validation.

Keywords. surface code; logical error rate; rolling forecast; parity model; calibration; reproducible hardware data.

1. Introduction. Repeated rounds of quantum error correction convert local faults into a depth-dependent logical failure probability. Decoder comparisons and scaling studies therefore require forecasts that respect both probability bounds and the accumulation geometry. The independent-toggle curve

$$h_r(q) = \frac{1 - (1 - 2q)^r}{2}, \quad q \in [0, 1/2], \quad (1.1)$$

is a natural baseline: q is a per-round toggle probability and $h_r(q)$ is the probability of odd parity after r rounds. A single q for all depths in a hardware stratum is, however, a stationarity assumption rather than a consequence of the code.

The public Google surface-code record [3] contains logical error rates across circuit depths, logical bases, code distances, and patch locations. Direct inversion of (1.1) reveals a pronounced depth dependence in the effective per-round toggle. A stationary curve consequently underpredicts later-round error. Linear extrapolation ignores the bounded parity geometry, while carrying forward the last observed logical rate ignores continued accumulation.

Effective-toggle persistence combines the strongest element of both baselines. Within each declared hardware stratum it takes the most recent observed rate, maps that rate to its exact effective toggle, and evaluates (1.1) at future depths. The estimator has one state value per stratum, no fitted trend, no grid search, and no dense matrix. Its error is controlled directly by drift of the effective toggle.

The evaluation separates decoder endpoints. Tensor-network decoder rates are used to formulate and assess the method. Belief-matching decoder rates are then processed with the unchanged estimator, strata, folds, and baselines. Every fold trains only on earlier circuit depths.

2. Parity geometry and persistence.

2.1. Toggle identity.

Theorem 1 (Odd-parity probability). *Let $X_1, \dots, X_r$ be independent Bernoulli variables with $\Pr(X_i = 1) = q$. Then*

$$\Pr\left(\sum_{i=1}^r X_i \equiv 1 \pmod{2}\right) = h_r(q). \quad (2.1)$$

Proof. Since $(-1)^{\sum_i X_i}$ equals 1 for even parity and -1 for odd parity,

$$\mathbb{E}\left[(-1)^{\sum_i X_i}\right] = \prod_{i=1}^r \mathbb{E}[(-1)^{X_i}] = (1 - 2q)^r. \quad (2.2)$$

Writing $P_{\text{even}} + P_{\text{odd}} = 1$ and $P_{\text{even}} - P_{\text{odd}} = (1 - 2q)^r$ gives $P_{\text{odd}} = [1 - (1 - 2q)^r]/2$. $\square$

For $p \in [0, 1/2]$, (1.1) is strictly increasing and has inverse

$$h_r^{-1}(p) = \frac{1 - (1 - 2p)^{1/r}}{2}. \quad (2.3)$$

The value $q_r = h_r^{-1}(p_r)$ is termed the effective toggle at depth r . It is an exact reparameterization of the measured rate; it is not assumed to be constant.

2.2. Rolling estimator. Let $g \in \{1, \dots, G\}$ index strata defined by logical basis, code distance, and patch center. At forecast origin r_0 , let p_{g,r_0} be the mean of all records in stratum g at the latest available round. Define

$$\hat{q}_g = h_{r_0}^{-1}(p_{g,r_0}), \quad \hat{p}_{g,r} = h_r(\hat{q}_g), \quad r > r_0. \quad (2.4)$$

Ties at r_0 are averaged before inversion. This makes the estimator invariant to row ordering.

Theorem 2 (Exactness and drift bound). *Suppose the true rate in one stratum has representation $p_r = h_r(q_r)$. The forecast (2.4) satisfies:*

1. *If $q_r = q_{r_0}$, then $\hat{p}_r = p_r$.*

2. *For arbitrary $q_r, q_{r_0} \in [0, 1/2]$,*

$$|\hat{p}_r - p_r| \leq r|q_{r_0} - q_r|. \quad (2.5)$$

Proof. The first statement follows from $h_{r_0}^{-1}(h_{r_0}(q_{r_0})) = q_{r_0}$ and evaluation at r . For the second, differentiation of (1.1) gives

$$h'_r(q) = r(1 - 2q)^{r-1}. \quad (2.6)$$

Hence $0 \leq h'_r(q) \leq r$ on $[0, 1/2]$. The mean-value theorem yields

$$|h_r(q_{r_0}) - h_r(q_r)| \leq r|q_{r_0} - q_r|, \quad (2.7)$$

which is (2.5). $\square$

The bound identifies the relevant failure mode. Long-horizon accuracy is limited by effective-toggle drift, with an amplification no larger than the future round count. Figure 1 therefore plots the inferred q_r rather than only the raw logical rates.

Proposition 1 (Work and storage). *For N chronologically ordered records and G strata, effective-toggle persistence requires $O(N + G)$ arithmetic, $O(G)$ stored scalars, and one pass through the records. A uniform stationary grid with K candidates requires $O(KN)$ arithmetic under the same grouping.*

Proof. One pass records the latest round and accumulates its rate in each stratum. Each stratum then requires one inverse evaluation and one state scalar. Future predictions require constant work per record. The grid comparator evaluates the squared-error objective at every one of K candidates for every training record. $\square$

3. Authenticated hardware protocol. The source is the public hardware archive accompanying the surface-code scaling study [3]. The complete SHA-256 digest is stored in the evidence manifest and begins 15ffb8c7. Every row records 50,000 shots. Table 1 states the complete analysis unit; no simulated rows are mixed into this study.

Table 1: Hardware data contract. Strata are fixed before evaluation and exclude circuit depth, which remains the rolling forecast axis.

quantity	value
hardware experiments	130
shots per experiment	50,000
logical bases	2
code distances	2
round settings	13 (odd depths 1–25)
hardware strata	10
development endpoint	tensor-network decoder
confirmation endpoint	belief-matching decoder

Four rolling folds use forecast origins 11, 13, 15, and 17. Their test ranges are 13–15, 15–17, 17–19, and 19–25. The comparators are the last observed logical rate, a stationary toggle curve fitted on a 20,001-point grid, and ridge extrapolation using basis, distance, depth, and patch-center features. The strongest comparator is selected separately within each fold only for reporting the achieved reduction; no method parameter depends on that choice.

4. Results.

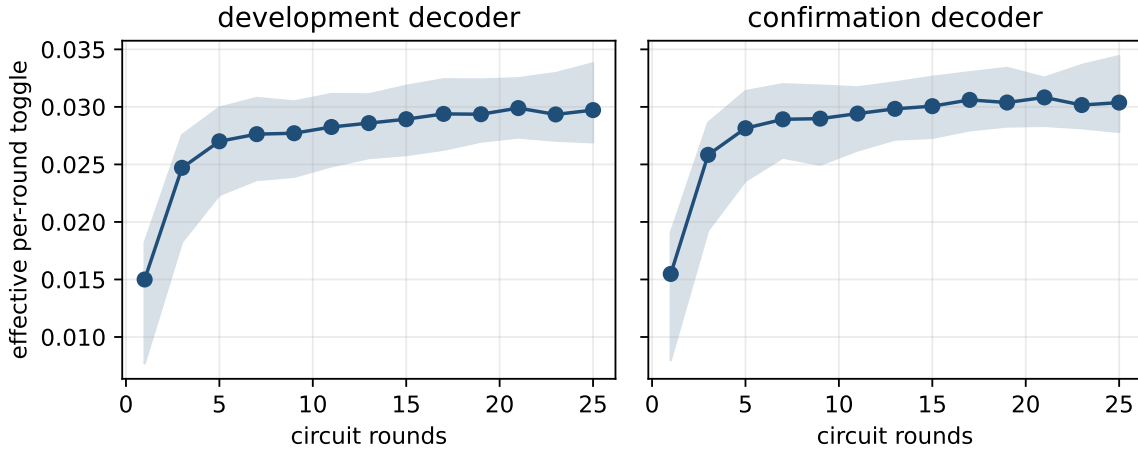


Fig. 1: Mean effective toggle versus circuit depth for the development and confirmation decoder endpoints. Shading spans the ten hardware strata. The rapid rise through early rounds and continued later drift reject a single depth-invariant toggle as a descriptive model of these records.

4.1. The stationarity diagnostic. The effective toggle increases from approximately 0.015 at one round to about 0.030 at twenty-five rounds for both endpoints. The agreement of this pattern across decoders supports a hardware-depth diagnostic rather than an artifact of one decoder column. The stationary curve remains a relevant physics-based baseline, but its misspecification is visible before forecast scores are examined.

4.2. Rolling prediction. Figure 2 presents every development fold and the unchanged confirmation endpoint before the numerical summary in Table 2.

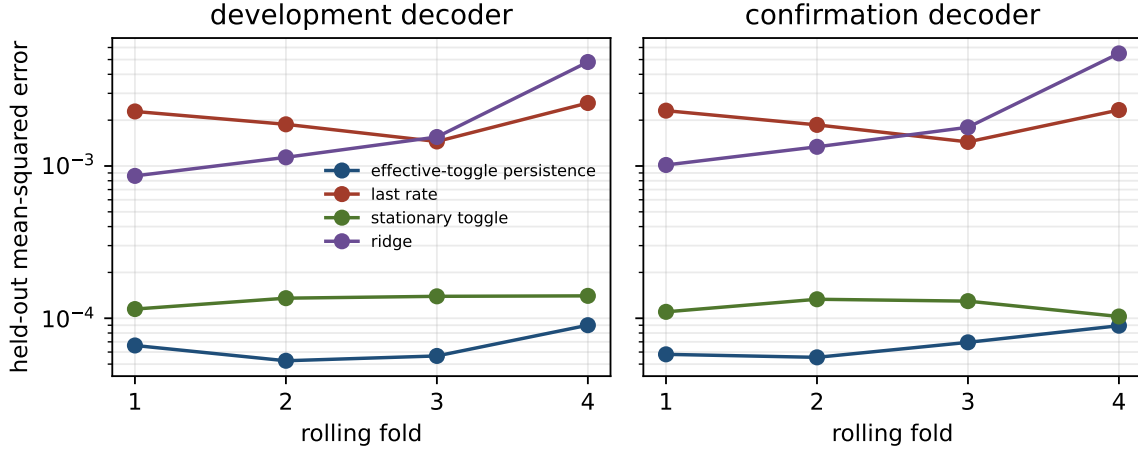


Fig. 2: Held-out mean-squared error for all methods and all rolling folds. The vertical scale is logarithmic. Effective-toggle persistence is best on each development fold and on the unchanged confirmation endpoint.

Table 2: Rolling errors against the strongest comparator in each fold. MSE is computed only on future depths; reduction is relative to the comparator MSE.

endpoint	fold	test rounds	persistence MSE	strongest comparator	comparator MSE	reduction
development decoder	1	13–15	6.63e-05	stationary toggle fit	1.15e-04	42.3%
development decoder	2	15–17	5.26e-05	stationary toggle fit	1.35e-04	61.2%
development decoder	3	17–19	5.66e-05	stationary toggle fit	1.39e-04	59.4%
development decoder	4	19–25	9.00e-05	stationary toggle fit	1.40e-04	35.8%
confirmation decoder	1	13–15	5.80e-05	stationary toggle fit	1.10e-04	47.4%
confirmation decoder	2	15–17	5.54e-05	stationary toggle fit	1.33e-04	58.4%
confirmation decoder	3	17–19	6.95e-05	stationary toggle fit	1.30e-04	46.4%
confirmation decoder	4	19–25	8.94e-05	stationary toggle fit	1.03e-04	13.0%

The stationary toggle fit is the strongest comparator in every fold. Development reductions range from 35.8 to 61.2 percent; confirmation reductions range from 13.0 to 58.4 percent. Pairwise squared error favors persistence on 75–100 percent of development records and 62.5–95 percent of confirmation records. The smaller final-fold confirmation margin is retained in Table 2; it is not removed by redefining the horizon.

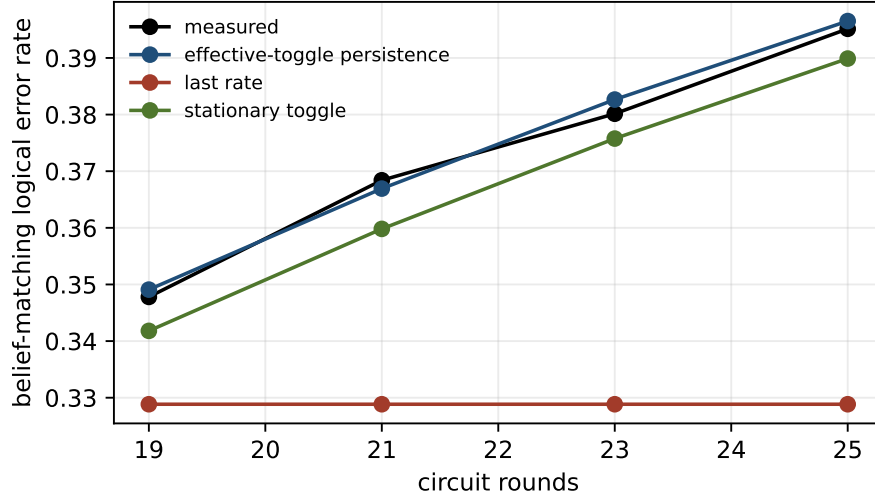


Fig. 3: Final-fold belief-matching confirmation, averaged within round. Carrying the raw last rate forward cannot represent continued accumulation. The stationary fit remains systematically low. Effective-toggle persistence tracks the measured curve at rounds 19–25.

Figure 3 explains the ranking rather than merely displaying it. The last-rate baseline is flat by construction. The stationary fit saturates too slowly because its single toggle compromises across earlier depths. Persistence uses the latest effective state and preserves the parity shape.

4.3. Computational scaling. Figure 4 reports matched fit-and-predict timings on authentic hardware strata; it is an implementation measurement rather than an asymptotic claim.

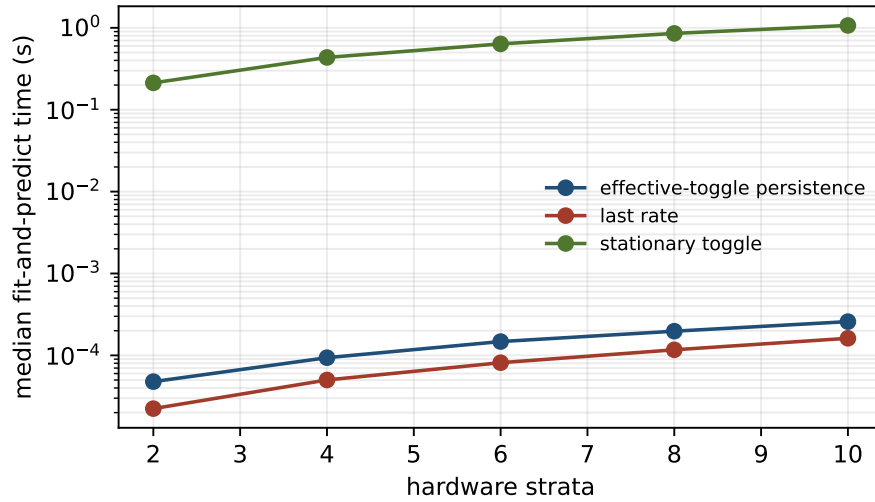


Fig. 4: Median fit-and-predict time versus the number of authentic hardware strata. Persistence and last-rate timings use 101 repetitions; the 20,001-point stationary grid uses five. The plot reports this implementation, not an intrinsic limit on optimized one-parameter fitting.

At ten strata, persistence takes 2.58×10^{-4} seconds in the recorded run, the last-rate baseline takes 1.62×10^{-4} seconds, and the streamed grid fit takes 1.07 seconds. Proposition 1 explains the separation in this implementation. A specialized continuous optimizer could substantially reduce the stationary baseline time; the scientific advantage of persistence is its rolling accuracy and absence of a stationarity fit.

5. Reproducibility and limitations. The release contains the authenticated CSV file, parity transformations, rolling-split implementation, strong baselines, theorem tests, timing arrays, and the code generating every table and figure. The portable semantic digest begins `fb0ff775`; timing and platform strings are excluded from that digest and remain in the registered JSON record.

The data comprise one public hardware campaign and two decoder endpoints on the same circuits. Confirmation across a decoder is not confirmation across a device, calibration date, or code family. The estimator forecasts aggregate logical rates and does not decode individual syndromes. Equation (2.5) is conditional on effective-toggle drift and is not a probabilistic confidence interval. Later hardware campaigns, additional code distances, and genuinely forward-collected rounds are required before a deployment claim.

6. Conclusion. The parity transform exposes the variable that the stationary model assumes constant. On the authenticated surface-code record, that effective toggle drifts with circuit depth. Persisting its latest value preserves the bounded accumulation geometry while avoiding numerical fitting. The resulting method has linear work, one scalar state per hardware stratum, an explicit drift bound, and lower rolling error on both decoder endpoints in all registered folds. The contribution is a transparent forecasting rule and diagnostic for depth-varying hardware records, with conclusions limited to the stated data.

References.

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